# Urban Vulnerability Mapping Using Satellite Images


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# Problem Definition

Urban authorities lack reliable methods to identify vulnerable urban areas from satellite images. This project proposes an AI-based system to classify areas into low, medium, and high vulnerability using physical and environmental features.

# Purpose and need

- Urban vulnerability arises from high building density, informal settlements, and environmental risks.
- Traditional methods rely on surveys and census data, which can be costly and outdated.
- These methods often lack fine spatial resolution and timely availability.
- Satellite imagery provides a scalable and consistent urban data source.
- AI-based analysis enables systematic identification of vulnerable urban areas.

# Project Objective

- The primary objective of the project is to spatially identify and classify urban vulnerability.
- Classification is based on absolute, predefined physical risk conditions.
- Each spatial unit is evaluated independently without reference to surrounding areas.
- Risk levels reflect directly observed vulnerability rather than relative comparison across regions.

# Literature Survey

**[1] Liang-Chieh Chen, Yukun Zhu, George Papandreou, Florian Schroff, and Hartwig Adam,**
**“Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation,”**
**Proceedings of the European Conference on Computer Vision (ECCV), 2018.**

Relevance:
This paper introduces DeepLabv3+, an advanced deep learning model for semantic image segmentation. It combines an encoder–decoder architecture with atrous (dilated) convolutions and Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale contextual information while preserving object boundaries. The model improves segmentation accuracy by refining object edges and reducing loss of spatial details. It widely used in applications like autonomous driving, medical imaging, and satellite image analysis.

# Literature Survey

**[2] X. Jean-Francois, A. Taubenböck, T. Esch, et al.,**
**“Urban Land Cover Classification Using Deep Neural Networks Based on VHR Multi-Spectral Image and Point Cloud Integration,” IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2024.**

Relevance:

This paper shows that using deep learning with high-resolution satellite images and 3D point cloud data gives better land cover results than traditional methods. It explains how combining different data sources helps clearly identify buildings, roads, and vegetation in cities

# Literature Survey

**[3] H. Taubenböck,**
**“Remote Sensing for the Analysis of Global Urbanization,” 2019.**

Relevance:

This work explains how satellite data, night-time light images, and socio-economic information can be combined to study how cities grow and change at different scales. It shows that using multiple data sources helps better understand urban growth patterns and internal city structures.

# Literature Survey

**[4] H. Taubenböck, et al.,**
**“Slums from Space—15 Years of Slum Mapping Using Remote Sensing,” Remote Sensing, 2016.**

Relevance:

Reviews how remote sensing techniques have been used over fifteen years to identify slum areas using high-resolution satellite imagery. It shows that features such as dense and irregular building patterns and poor road structures can help detect informal settlements from space. This study is relevant because it demonstrates how satellite imagery can be used to identify vulnerable urban areas, supporting urban vulnerability assessment and planning.

# Literature Survey

**[5] S. Chakraborty, et al.,**
**“Mapping Urban Heat Vulnerability Using Remote Sensing and GIS,” 2019.**

Relevance:

This study uses GIS and satellite data along with population information to map areas that are most vulnerable to urban heat. It shows that combining heat intensity with population characteristics helps identify high-risk heat hotspots in cities.

# Literature Survey

**[6] B. A. Thiam, et al.,**
**“Deep Learning-based Land Use and Land Cover Changes Detection from Satellite Imagery: A Case Study of the City of Richard Toll,” 2024.**

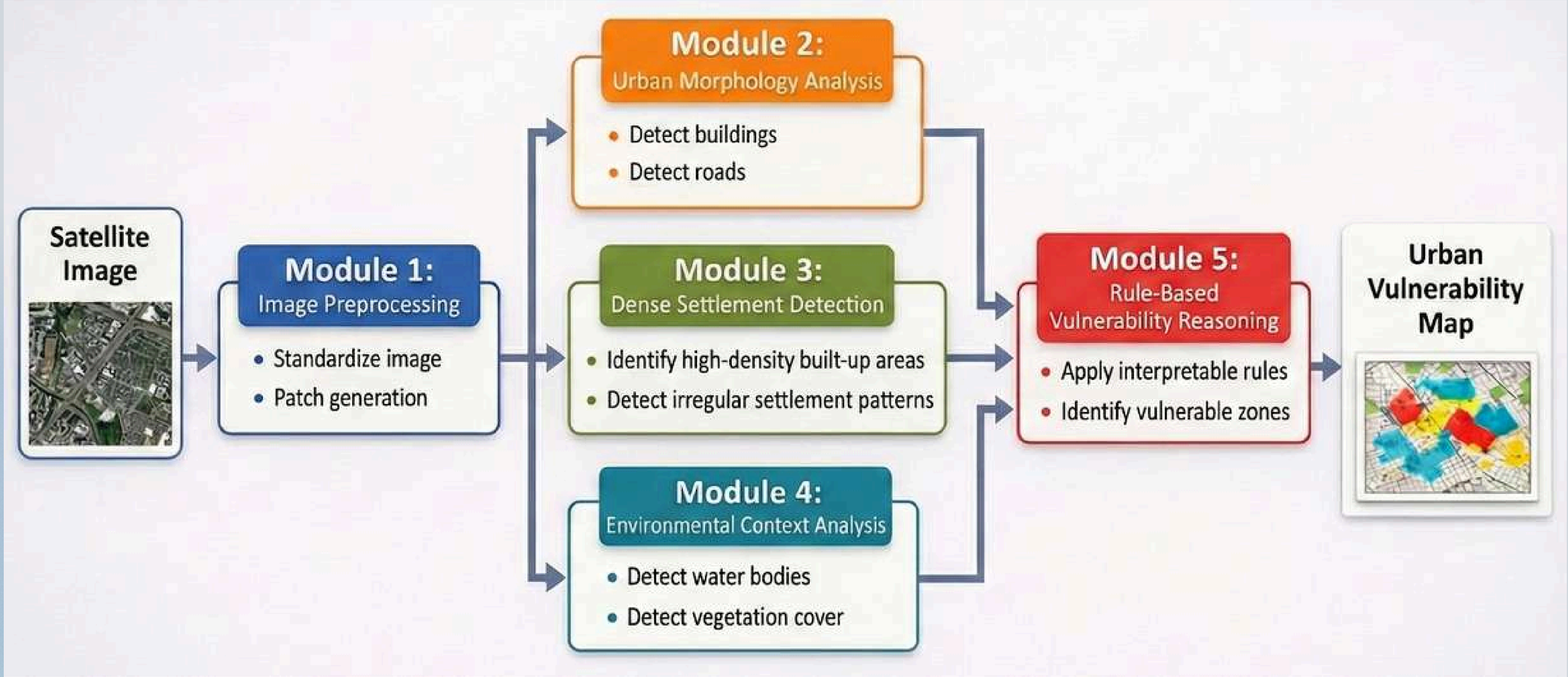
Relevance:

This study uses deep learning models like U-Net on satellite images from different years to detect land use and land cover changes. It shows that U-Net gives better and more detailed change maps, helping in automatic monitoring of urban and agricultural areas.

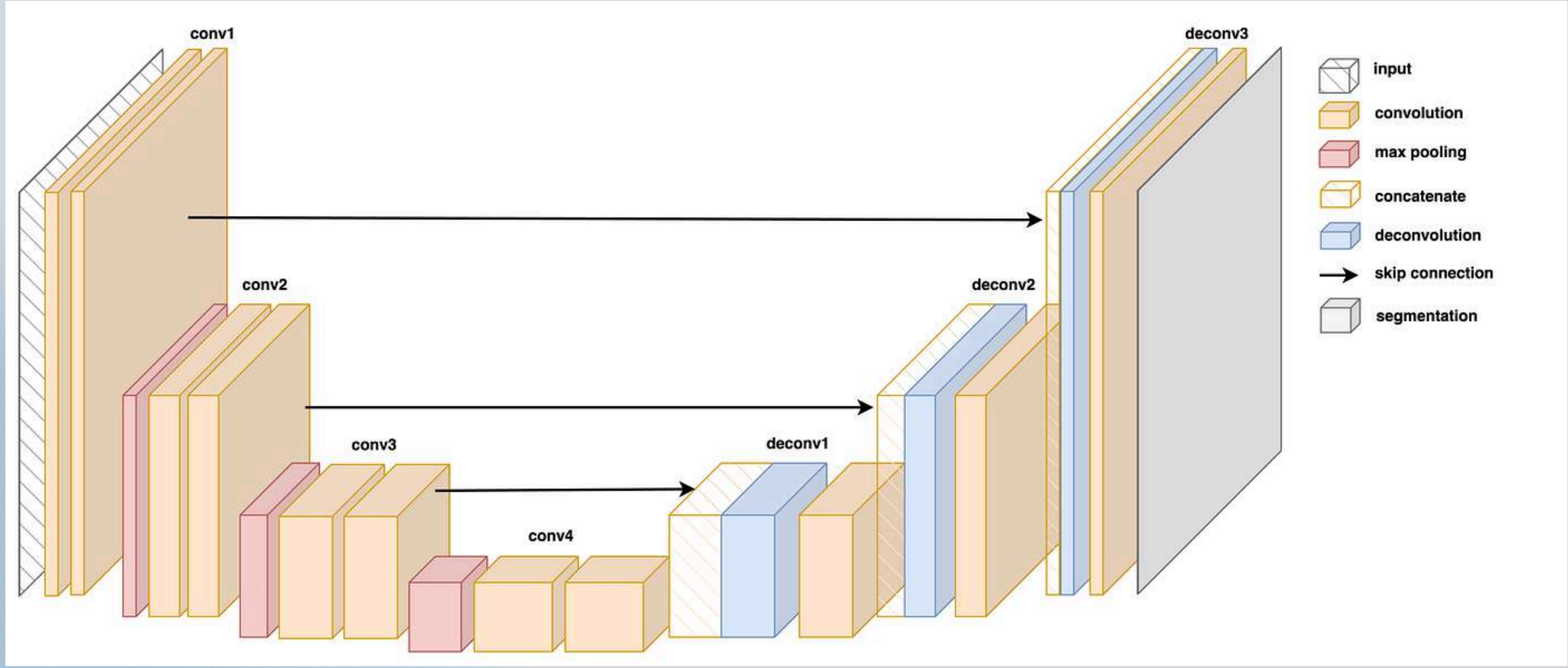
# Literature Survey

| Paper Title | Relevance | Pros | Cons |
|---|---|---|---|
| **Urban Land Cover Classification Using Deep Neural Networks Based on VHR Multi-Spectral Image and Point Cloud Integration (2024)** | Shows that deep learning combined with high-resolution satellite images and point cloud data improves urban land cover classification. | • High accuracy compared to traditional OBIA• Uses both spectral and 3D information• Good performance in complex urban areas | • Requires point cloud data, which is not always available• Higher computational cost |
| **Remote Sensing for the Analysis of Global Urbanization (2019)** | Explains how multiple remote sensing and socio-economic datasets can be used to study global and local urban growth patterns. | • Covers urbanization at multiple scales• Integrates social and physical data• Strong conceptual frameworks | • No single algorithm or accuracy metrics• More descriptive than implementation-focused |
| **Slums from Space—15 Years of Slum Mapping Using Remote Sensing (2016)** | Reviews 15 years of methods for mapping slums using satellite imagery and highlights effective techniques. | • Comprehensive review of methods• Identifies best-performing techniques• Useful for understanding challenges | • No new model proposed• Limited focus on real-time or large-scale automation |
| **Mapping Urban Heat Vulnerability Using Remote Sensing and GIS (2019)** | Demonstrates how GIS and satellite data combined with population data can identify urban heat risk zones. | • Combines physical and social factors• Useful for urban planning• Spatially detailed vulnerability maps | • No machine learning comparison• Results depend heavily on indicator selection |
| **Deep Learning-based Land Use and Land Cover Changes Detection from Satellite Imagery (2024)** | Uses deep learning segmentation (U-Net) on multi-temporal images to detect land cover changes automatically. | • Automated change detection• Detailed class-wise results• U-Net performs better than FCN-8 | • Accuracy metrics not clearly reported• Limited to a single study area |

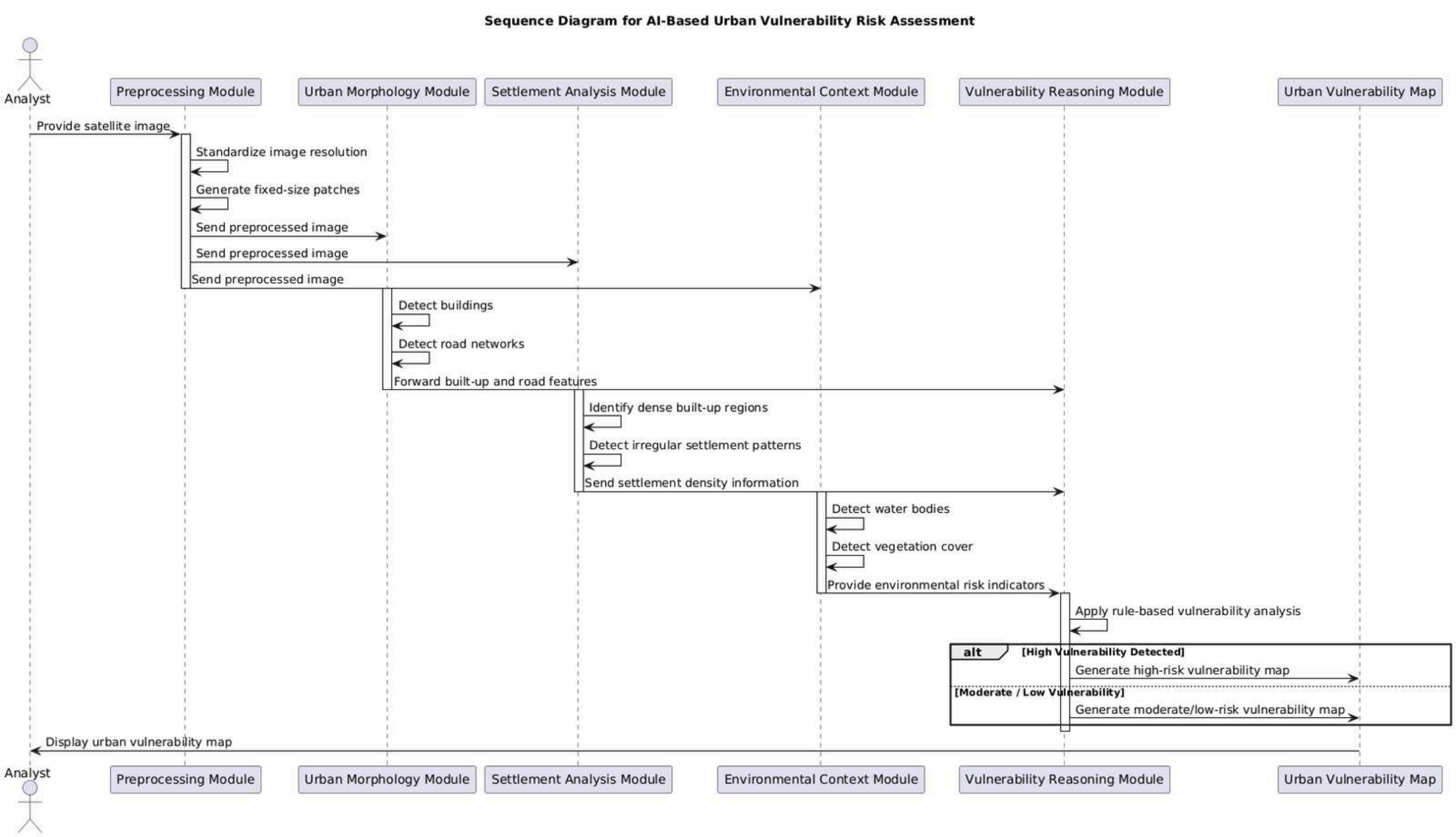
# Architecture Diagram



# Unet Architecture Diagram



# Sequence Diagram



# Modules

**Module 1: Image Preprocessing**

Goal: Prepare satellite imagery for analysis

- Standardize image format: Resize and normalize satellite images to ensure uniform input
- Patch generation: Divide large satellite images into smaller patches (e.g., 256×256px) to simplify processing
- This enables efficient processing by deep learning models and downstream tasks

# Modules

**Module 2: Urban Morphology Analysis**

Goal: Identify built environment structure

- Detect buildings: A U-Net based deep learning model was trained to perform semantic segmentation for automatic building footprint detection from satellite imagery.
- Detect roads: Used DeepLabV3+ semantic segmentation model to detect road networks from satellite images.
- These elements help understand urban form and accessibility
- ***Dataset:*** *SpaceNet Dataset (specifically the AOI_2_Vegas subset).*
- ***Dataset:*** *DeepGlobe Road Extraction Dataset.*

# Modules

**Module 3: Dense Settlement Detection**

Goal: Detect informal/slum-like regions

- Slum segmentation: Apply a deep learning–based Unet semantic segmentation model to detect regions exhibiting characteristics of informal settlements.
- Output: Generate binary masks highlighting potential slum areas, which are used for further urban vulnerability analysis.

- ***Dataset:*** *OpenCity*

# Modules

**Module 4: Environmental Context Analysis**

Goal: Assess environmental risks

- Water bodies: Detect proximity to rivers/lakes to assess flood risk
- Vegetation cover: robust U-Net model was trained on automatically generated satellite datasets using vegetation masking to accurately segment vegetation regions.
- ***Dataset:*** *Esri ArcGIS World Imagery*
- ***Dataset:*** *Sentinel-2 Water Bodies Dataset*

# Modules

## Module 5: Rule-Based Vulnerability Reasoning

- Apply spatial risk rules
- Calculate vulnerability indicators
- Computed metrics:
  - Built-up ratio
  - Slum density
  - Road deficiency
  - Flood exposure risk
  - Vegetation buffer
  - Spatial entropy
- Risk Score Formula:
- Risk Score =
  - (Slum Density × 60)+ (Entropy × 20)+ (Road Deficit × 15)+ (Flood Exposure × 25)- (Vegetation Buffer × 10)

## Risk classification:

- High (75–100)
- Medium (50-75)
- Low (<50)

# Modules

## Module 5: Rule-Based Vulnerability Reasoning

- Urban Vulnerability Map
- Outputs generated:
  - Combined overlay map
  - Layer maps:
    - Built-up
    - Roads
    - Water
    - Vegetation
    - Slum areas
  - Global vulnerability score
  - Patch-wise telemetry
  - LLM risk analysis report

# Requirements

1. ***Hardware Requirements***

- Minimum 12 GB RAM for image processing and model execution
- CPU Intel Core i5 / AMD Ryzen 5 or higher.
- NVIDIA GPU with CUDA support (≥ 6 GB VRAM recommended)
- Minimum 50 GB free disk space (SSD preferred)

2. ***Software Requirements***

- Windows 10 / 11 (64-bit) or Ubuntu 20.04 or later
- Python 3.x development environment
- TensorFlow for AI model development
- OpenCV for satellite image processing
- NumPy for feature computation
- GIS tools for spatial visualization

# Expected Outcome

1. Automated detection of built-up areas and road networks from satellite imagery
2. Identification of slum regions based on spatial features
3. Integration of environmental factors such as vegetation and water bodies
4. Generation of urban vulnerability maps highlighting high, moderate, and low-risk zones

# Conclusion

This project presents a transparent AI-based approach for assessing urban vulnerability using satellite imagery. By analyzing physical urban patterns and environmental indicators at a patch level, the system classifies regions into low, medium, and high risk using clear rule-based logic. The proposed solution supports urban analysis and prioritization while remaining interpretable, scalable, and suitable for decision-support applications.

# References

[1]Chen, L. C., Zhu, Y., Papandreou, G., Schroff, F., & Adam, H. (2018). Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation. Proceedings of the European Conference on Computer Vision (ECCV 2018), pp. 833–851. doi: https://doi.org/10.1007/978-3-030-01234-2_49

[2]Fawzy, M., Taubenböck, H., Esch, T., et al. (2024). Urban Land Cover Classification Using Deep Neural Networks Based on VHR Multi-Spectral Image and Point Cloud Integration. Proceedings of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2024). 5377–5381.

[3]Taubenböck, H. (2019). Remote Sensing for the Analysis of Global Urbanization. Remote Sensing of Environment, Elsevier. researchgate.net/publication/334318755_Remote_Sensing_for_the_Analysis_of_Global_Urbanization

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# Thank You